

BS – 101 Engineering Physics

Lecture – 16
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Rectifier

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- Because of their ability to conduct current in one direction and block current in the other direction, diodes are used in circuits called rectifiers that convert ac voltage into dc voltage.
- Rectifiers are found in all dc power supplies that operate from an ac voltage source. A power supply is an essential part of each electronic system from the simplest to the most complex.



The Basic DC Power Supply

- All active electronic devices require a source of constant dc that can be supplied by a battery or a dc power supply.
- The **dc power supply** converts the standard 220 V, 50 Hz ac voltage available at wall outlets into a constant dc voltage.
- The dc power supply is one of the most common circuits you will find, so it is important to understand how it works.
- The voltage produced is used to power all types of electronic circuits including consumer electronics (televisions, DVDs, etc.), computers, industrial controllers, and most laboratory instrumentation systems and equipment.

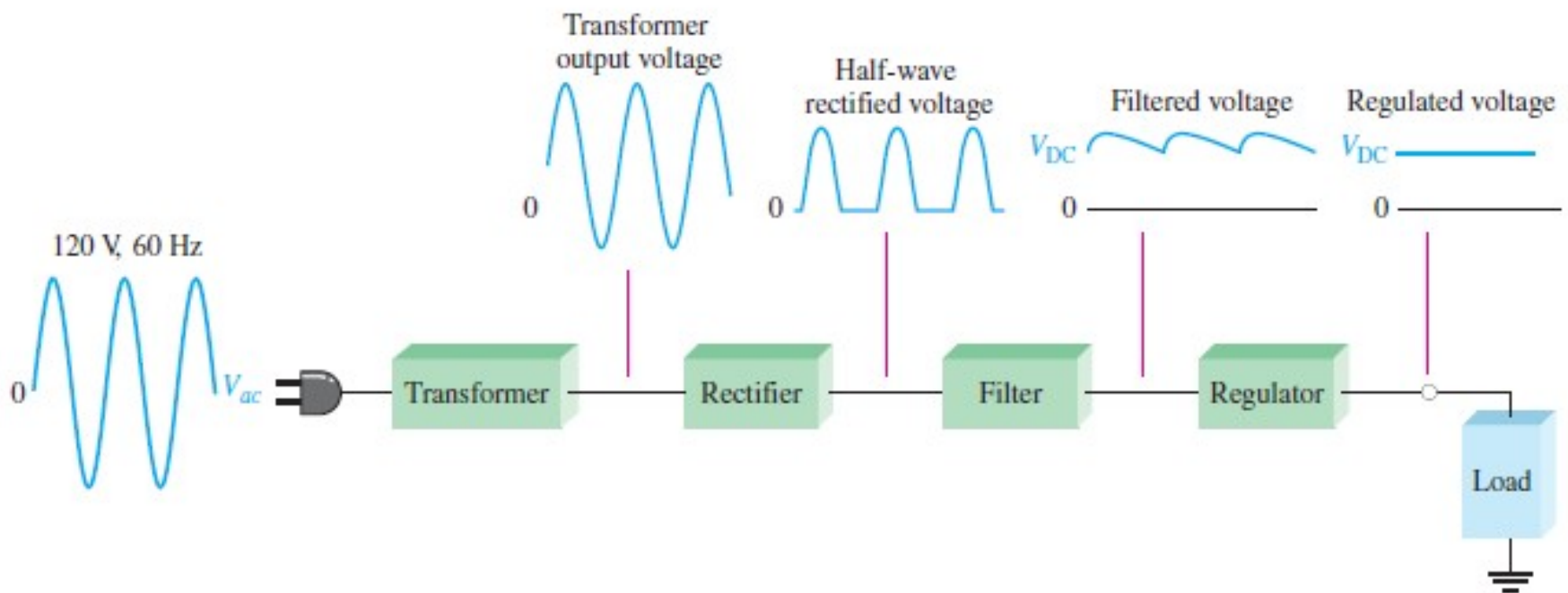


The Basic DC Power Supply

(Contd.)

- The basic block diagram of the complete power supply is shown in figure on next slide.
- Generally the ac input line voltage is stepped down to a lower ac voltage with a transformer.
- The rectifier converts the ac input voltage to a pulsating dc voltage.
- The rectifier can be either a halfwave rectifier or a full wave rectifier.
- The filter eliminates the fluctuations in the rectified voltage and produces a relatively smooth dc voltage.
- The regulator is a circuit that maintains a constant dc voltage for variations in the input line voltage or in the load.
- The load is a circuit or device connected to the output of the power supply and operates from the power supply voltage

The Basic DC Power Supply (Contd.)

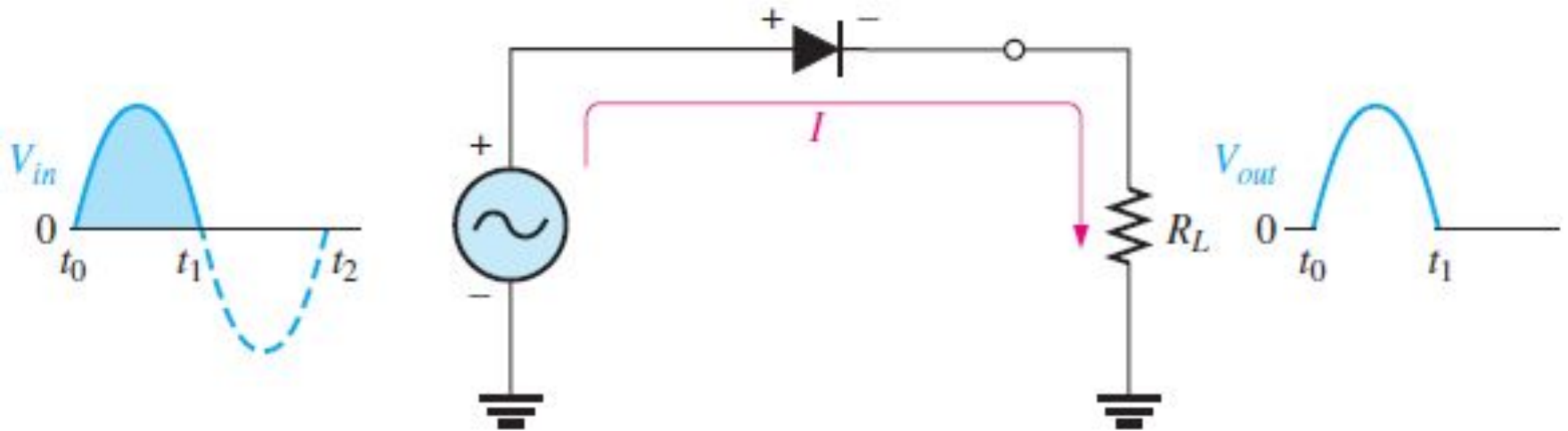


Half Wave Rectifier

Operation

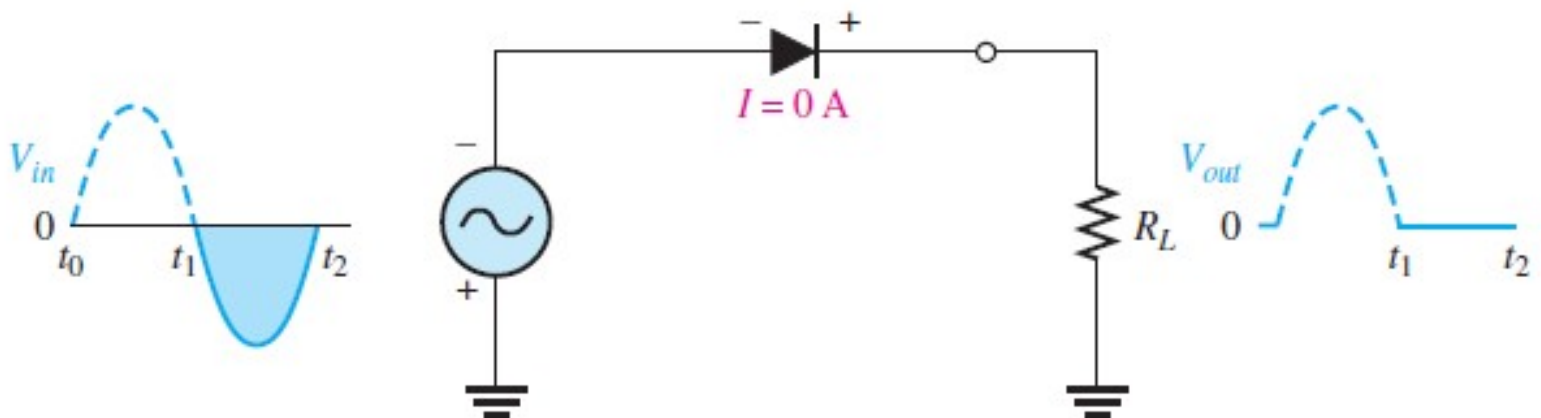
- Figure on next slide illustrates the process called *half-wave rectification*.
- A diode is connected to an ac source and to a load resistor, R_L , forming a **half-wave rectifier**.
- Keep in mind that all ground symbols represent the same point electrically.
- Let's examine what happens during one cycle of the input voltage using the ideal model for the diode.
- When the sinusoidal input voltage (V_{in}) goes positive, the diode is forward-biased and conducts current through the load resistor.
- The current produces an output voltage across the load R_L , which has the same shape as the positive half-cycle of the input

Half Wave Rectifier Operation (Contd.)



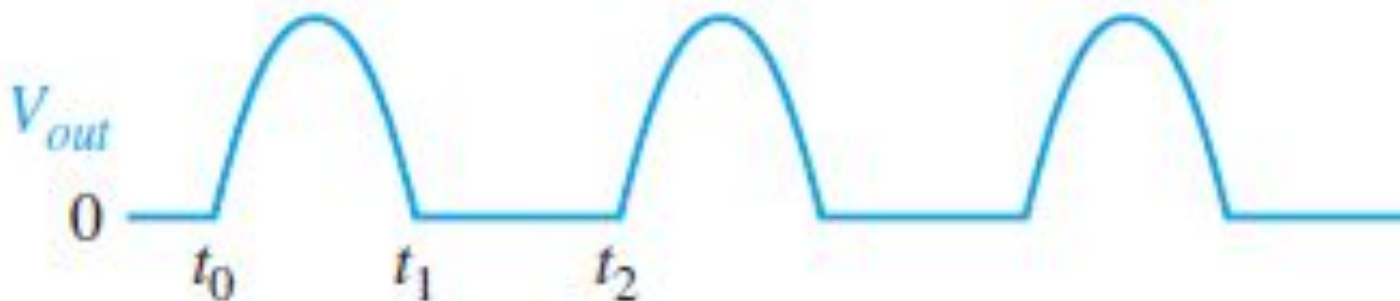
Half Wave Rectifier Operation (Contd.)

- When the input voltage goes negative during the second half of its cycle, the diode is reverse-biased. There is no current, so the voltage across the load resistor is 0 V, as shown in Figure.



Half Wave Rectifier Operation (Contd.)

- The net result is that only the positive half-cycles of the ac input voltage appear across the load. Since the output does not change polarity, it is a pulsating dc voltage with a frequency of 50 Hz.



Average Value of the Half Wave Output Voltage

- The average value of the half-wave rectified output voltage is the value you would measure on a dc voltmeter.
 - Mathematically, it is determined by finding the area under the curve over a full cycle, and then dividing by the number of radians in a full cycle.
 - The result of this is expressed in Equation, where V_P is the peak value for the voltage. This equation shows that the average value is approximately 31.8% of V_P for a half-wave rectified voltage.
- www.electronics-tutorials.com/floyd

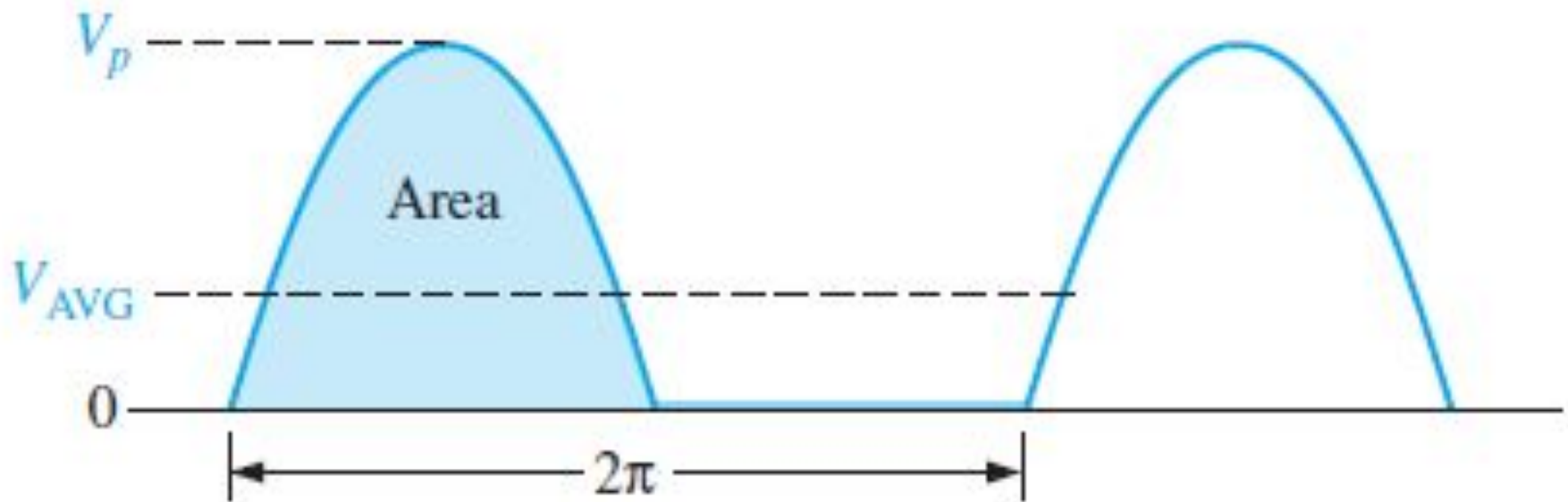
Average Value of the Half Wave Output Voltage (Contd.)

- The average value of a half-wave rectified sine wave is the area under the curve divided by the period (2π). The equation for a sine wave is

$$V_{AVG} = \frac{\text{area}}{2\pi} = \frac{V_P}{2\pi} \int_0^{\pi} \sin \theta d\theta = \frac{V_P}{2\pi} (-\cos \theta) \Big|_0^{\pi}$$

- $V_{AVG} = \frac{V_P}{\pi}$

Average Value of the Half Wave Output Voltage (Contd.)



Example

- What is the average value of the half-wave rectified voltage in Figure





Example (Contd.)

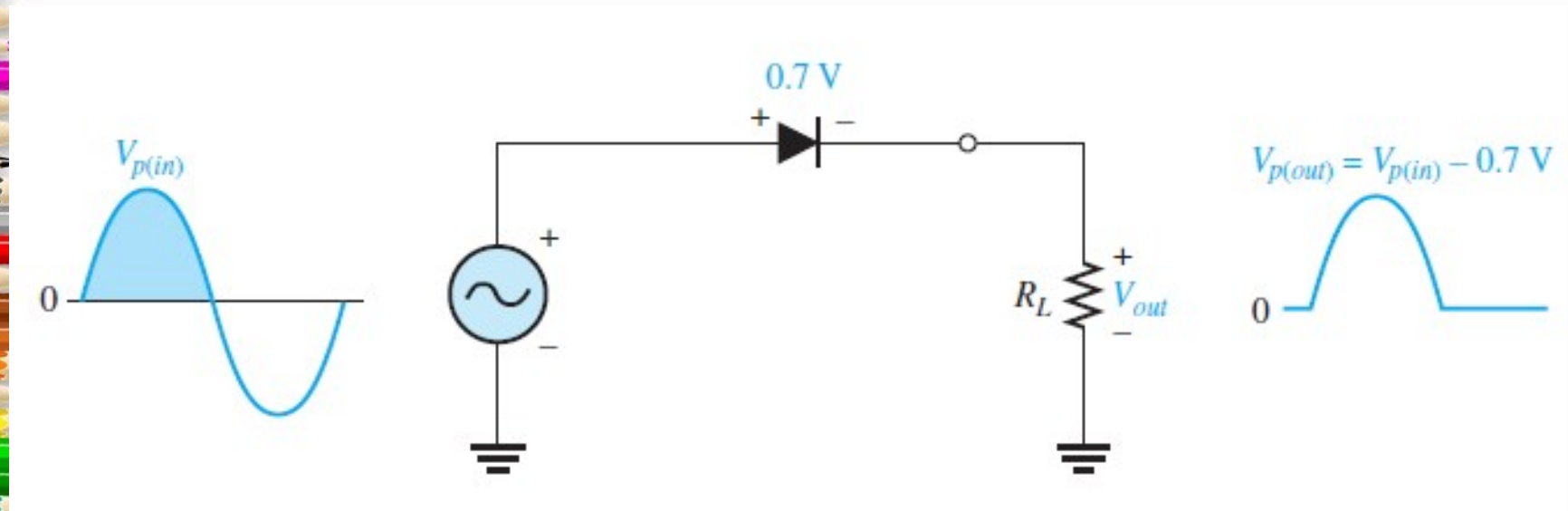
- Determine the average value of the half-wave voltage if its peak amplitude is 12 V.

Effect of the Barrier Potential on the Half Wave Rectifier

Output

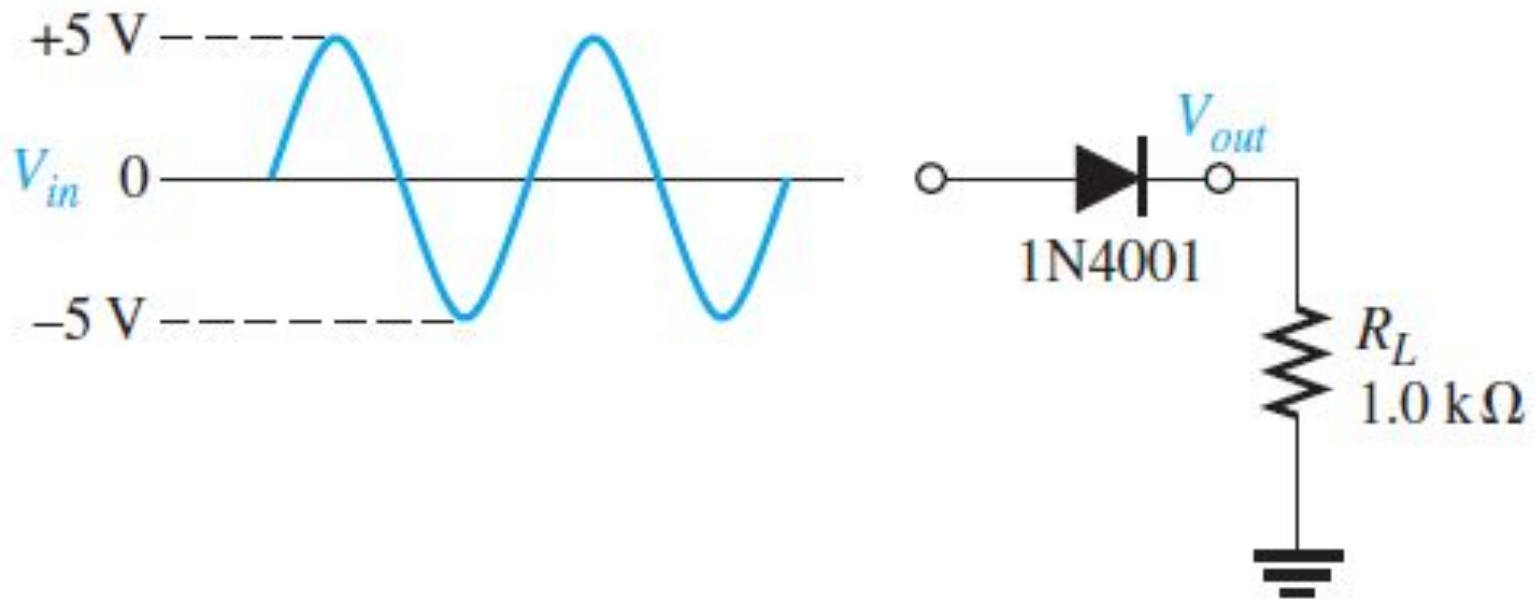
- In the previous discussion, the diode was considered ideal. When the practical diode model is used with the barrier potential of 0.7 V taken into account, this is what happens.
- During the positive half-cycle, the input voltage must overcome the barrier potential before the diode becomes forward-biased.
- This results in a half-wave output with a peak value that is 0.7 V less than the peak value of the input. The expression for the peak output voltage is
$$V_{P(out)} = V_{P(in)} - 0.7V$$

Effect of the Barrier Potential on the Half Wave Rectifier Output (Contd.)



Example

- Draw the output voltage of the rectifier as shown in figure.



Example (Contd.)

- Draw the output voltage of the rectifier as shown in figure.

